

Report On FoodCorps. Market Analysis in the stores across UK.

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1. Executive Summary:

This report presents a detailed analysis of FoodCorp's regional store performance to support the management in identifying the most promising store for increased marketing investment. FoodCorp, a growing company with four stores across London, Birmingham, and Nottingham, seeks to determine which store offers the best growth opportunities for increased marketing investment. This analysis evaluates store performance, customer behaviour, product trends, and city-level insights.

2. Data Cleaning & Preparation Steps:

The data was cleaned and pre-processed to ensure accuracy and consistency for analysis. Key steps discussed below-

- 1) Removing duplicates for customer, receipt and product details.
- 2) Handling missing values through imputation or flagging for the values like customer id, store code, product details.
- 3) Standardizing formats for addresses and postcode to easily get the city column.
- 4) Errors such as outliers and inconsistencies for were corrected for value and quantity measures.
- 5) When analysing data for a business like FoodCorp, it is needed to organize the data in a way that facilitates efficient querying, insightful analysis, and intuitive reporting. This is achieved by designing a data warehouse schema using fact and dimension tables.
- 6) In the 'stores_dim' created as dimension table, a new column called 'Store Location' was added to uniquely identify store names. It will replace the store codes with combination of city and first-part of postalcode like 'London EC3V' to easily identify and analyse.

Section 1: The KPIs For Analysis

The KPIs used for the Analysis are divided into Sales performance, customer behaviour, Product performance and Overall Store Performance Trends.

Sales performance KPIs -

KPI 1: Total Sales for Each Stores Per Quarter

- a) **Purpose:** Total Sales made by Customers in each store for a given period.
- b) **Formula:** Sum of Value , Group By & Order By -store location, year & Quarter.
- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.
- d) **Code:**

```
SELECT
    sd.store_location,td.year,
    CONCAT('Q', CEIL(MONTH(sf.purchased_at) / 3)) AS quarter,
    SUM(sf.value) AS total_sales
FROM sales_fact sf
JOIN stores_dim sd ON sf.store_code = sd.store_code
JOIN time_dim td ON sf.purchased_at = td.purchased_at
GROUP BY sd.store_location, td.year, quarter
ORDER BY sd.store_location, td.year, quarter;
```

- e) **Reference:** Extracted output has been visualized in fig 1 below.
- f) **Additional Notes:** Quarters are represented by concatenating 'Q' with quarter number followed by year.

KPI 2: Sales Rate Per Day of Week

- a) **Purpose:** Proportion of sales acquired by each store location in a weekday.
- b) **Formula:** Sum of Value (Total Daily Sales)/ sum of Total sales across the store.
- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.
- d) **Code:**

```
WITH daily_sales AS (
  SELECT s.store_location,
  EXTRACT(DAYOFWEEK FROM sf.purchased_at) AS day_of_week,
  SUM(sf.value) AS total_sales
  FROM sales_fact sf
  JOIN stores_dim s ON sf.store_code = s.store_code
  GROUP BY s.store_location, EXTRACT(DAYOFWEEK FROM sf.purchased_at)
),
-- Calculated total sales per store location and day of the week above.
total_sales_per_store AS (
  SELECT store_location,
  SUM(total_sales) AS overall_sales
  FROM daily_sales
  GROUP BY store_location
)
-- Calculated total sales per store location across all days above.
SELECT d.store_location,
CASE
  WHEN d.day_of_week = 1 THEN 'Sunday'
  WHEN d.day_of_week = 2 THEN 'Monday'
  WHEN d.day_of_week = 3 THEN 'Tuesday'
  WHEN d.day_of_week = 4 THEN 'Wednesday'
  WHEN d.day_of_week = 5 THEN 'Thursday'
  WHEN d.day_of_week = 6 THEN 'Friday'
  WHEN d.day_of_week = 7 THEN 'Saturday'
END AS day_name, d.total_sales,
ROUND((d.total_sales / ts.overall_sales) * 100, 2) AS sales_proportion
FROM daily_sales d
JOIN total_sales_per_store ts ON d.store_location = ts.store_location
ORDER BY d.store_location,
CASE
  WHEN d.day_of_week = 2 THEN 1 -- Monday
  WHEN d.day_of_week = 3 THEN 2 -- Tuesday
  WHEN d.day_of_week = 4 THEN 3 -- Wednesday
  WHEN d.day_of_week = 5 THEN 4 -- Thursday
  WHEN d.day_of_week = 6 THEN 5 -- Friday
  WHEN d.day_of_week = 7 THEN 6 -- Saturday
  WHEN d.day_of_week = 1 THEN 7 -- Sunday
END;
END;
```

- e) **Reference:** Extracted output has been visualized in fig 2 below.
- f) **Additional Notes:** The percentage of daily says is represented as the ranking of a weekdays starting from Sunday to Saturday.

Customer Behaviour Metrics:

KPI 3: New Customers/ First Time Buyers.

- a) **Purpose:** to calculate new customers per quarter in each store.
- b) **Formula:** Min Purchased at per quarter to calculate new customers, count of customer id.
- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.

d) Code:

```
WITH new_customers AS (
  SELECT sf.customer_id, sd.store_location,
  MIN(sf.purchased_at) AS first_purchase_date
  FROM sales_fact sf
  JOIN stores_dim sd ON sf.store_code = sd.store_code
  GROUP BY sf.customer_id, sd.store_location
),
quarterly_new_customers AS (
  SELECT store_location,
  CONCAT('Q', EXTRACT(QUARTER FROM first_purchase_date)) AS quarter,
  EXTRACT(YEAR FROM first_purchase_date) AS year,
  COUNT(DISTINCT customer_id) AS New_Customer_Count
  FROM new_customers
  GROUP BY store_location, EXTRACT(QUARTER FROM first_purchase_date), EXTRACT(YEAR FROM first_purchase_date)
)
SELECT store_location, quarter, year, New_Customer_Count
FROM quarterly_new_customers
ORDER BY store_location, year, quarter;
```

▶ (4) Spark Jobs

Table	store_location	quarter	year	New_Customer_Count
1	Birmingham Rd	Q1	2020	270

e) **Reference:** Extracted output has been visualized in fig 3 below.

f) **Additional Notes:** If the first purchase occur in a quarter, we take the count of distinct customer id as the number of new customers.

KPI 4: Customer Lifetime Value (CLV)

a) **Purpose:** Quantifies customer value over their relationship with the store.

b) **Formula:** Average total revenue per customer per quarter, per store within their respective first and last purchase date.

c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.

d) Code:

```
WITH customer_lifetime_revenue AS (
  SELECT
  sf.customer_id,
  sd.store_location,
  SUM(sf.value) AS total_revenue_per_customer,
  MIN(sf.purchased_at) AS first_purchase_date,
  MAX(sf.purchased_at) AS last_purchase_date
  FROM sales_fact sf
  JOIN stores_dim sd ON sf.store_code = sd.store_code
  GROUP BY sf.customer_id, sd.store_location),
customer_lifetime AS (
  SELECT store_location, customer_id, total_revenue_per_customer,
  DATEDIFF(last_purchase_date, first_purchase_date) AS customer_lifetime_days
  FROM customer_lifetime_revenue
)
SELECT store_location,
ROUND(AVG(total_revenue_per_customer), 2) AS avg_clv
FROM customer_lifetime
GROUP BY store_location
ORDER BY avg_clv DESC;
```

- e) **Reference:** Extracted output has been visualized in fig 4 below.
- f) **Additional Notes:** A mean value is calculated as a reference for the whole customer base per store.

KPI 5: Cohort Retention Analysis

- a) **Purpose:** It tracks how well the business retains its customer base per cohort, per store.
- b) **Formula:** (retained customer base/cohort size)*100, per store, per quarter.
- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.
- d) **Code:**

```
-- Find the first purchase date for each customer
WITH first_purchase AS (
  SELECT r.customer_id,s.store_location,
  MIN(CONCAT(EXTRACT(YEAR FROM t.purchased_at), ' Q', EXTRACT(QUARTER FROM t.purchased_at))) AS cohort
  FROM receipts_dim r
  JOIN stores_dim s ON r.store_code = s.store_code
  JOIN time_dim t ON r.purchased_at = t.purchased_at
  GROUP BY r.customer_id, s.store_location
),
-- Track purchases by each customer per quarter-year
quarterly_purchases AS (
  SELECT r.customer_id,s.store_location,
  CONCAT(EXTRACT(YEAR FROM t.purchased_at), ' Q', EXTRACT(QUARTER FROM t.purchased_at)) AS year_quarter
  FROM receipts_dim r
  JOIN stores_dim s ON r.store_code = s.store_code
  JOIN time_dim t ON r.purchased_at = t.purchased_at
  GROUP BY r.customer_id, s.store_location, EXTRACT(YEAR FROM t.purchased_at), EXTRACT(QUARTER FROM t.purchased_at)
),
-- Join first purchases with quarterly purchases to track retention
cohort_retention AS (
  SELECT fp.store_location,fp.cohort,qp.year_quarter,
  COUNT(DISTINCT qp.customer_id) AS retained_customers
  FROM first_purchase fp
  JOIN quarterly_purchases qp
  ON fp.customer_id = qp.customer_id
  AND fp.store_location = qp.store_location
  GROUP BY fp.store_location, fp.cohort, qp.year_quarter
),
-- Get the total cohort size (number of customers) per store and cohort
total_cohort_size AS (
  SELECT store_location,cohort,
  COUNT(DISTINCT customer_id) AS cohort_size
  FROM first_purchase
  GROUP BY store_location, cohort
)
-- Calculate retention rate by dividing retained customers by the cohort size
SELECT c.store_location,c.cohort,c.year_quarter,c.retained_customers,tc.cohort_size,
ROUND(100.0 * c.retained_customers / tc.cohort_size, 2) AS retention_rate
FROM cohort_retention c
JOIN total_cohort_size tc ON c.store_location = tc.store_location AND c.cohort = tc.cohort
ORDER BY c.store_location, c.cohort, c.year_quarter;
```

- e) **Reference:** Extracted output has been visualized in fig 5 below.
- f) **Additional Notes:** Cohort size is basically the total number of customers. It's calculating retention rate by dividing retained customers by the cohort size.

Product Performance Metrics –

KPI 6: Top Five Selling Department.

- a) **Purpose:** It identifies which department made most of the sales revenue.
- b) **Formula:** Total sales value per department per department per quarter.

- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.
- d) **Code:**

```
WITH sales_by_department AS (
    SELECT s.store_location,
           CONCAT(EXTRACT(YEAR FROM t.purchased_at), ' Q', EXTRACT(QUARTER FROM t.purchased_at)) AS year_quarter,
           p.department_name,
           SUM(f.value) AS total_revenue
    FROM sales_fact f
    JOIN stores_dim s ON f.store_code = s.store_code
    JOIN time_dim t ON f.purchased_at = t.purchased_at
    JOIN products_dim p ON f.product_code = p.product_code
    GROUP BY s.store_location, year_quarter, p.department_name
),
ranked_departments AS (
    SELECT store_location, year_quarter, department_name, total_revenue,
           RANK() OVER (PARTITION BY store_location, year_quarter ORDER BY total_revenue DESC) AS rank
    FROM sales_by_department
)
SELECT store_location, year_quarter, department_name, total_revenue
FROM ranked_departments
WHERE rank <= 5
ORDER BY store_location, year_quarter, rank;
```

- e) **Reference:** Extracted output has been visualized in fig 6 below.
- f) **Additional Notes:** Departments are represented by the percentage of total sales per store, every year in graph.

Store Performance Metrics –

KPI 6: Store Performance Trend Over Time.

- a) **Purpose:** It gives overall store performance per store location, per year-quarter.
- b) **Formula:** Total quantity sold, total transactions, total revenue per quarter, per stores.
- c) **Steps to realize the KPI:** Codes to calculate the KPI pasted below, extracted output has been analysed through data visualization, it will be discussed in second section.
- d) **Code:**

```
WITH quarterly_sales AS (
    SELECT s.store_location,
           EXTRACT(QUARTER FROM t.purchased_at) AS quarter,
           EXTRACT(YEAR FROM t.purchased_at) AS year,
           SUM(f.value) AS total_revenue,
           SUM(f.value)/COUNT(DISTINCT f.receipt_id) AS avg_transaction_value,
           SUM(f.qty) AS total_quantity_sold,
           COUNT(DISTINCT f.receipt_id) AS total_transactions
    FROM sales_fact f
    JOIN stores_dim s ON f.store_code = s.store_code
    JOIN time_dim t ON f.purchased_at = t.purchased_at
    GROUP BY s.store_location, EXTRACT(QUARTER FROM t.purchased_at), EXTRACT(YEAR FROM t.purchased_at)
)
SELECT store_location, year,
       CONCAT('Q', quarter) AS quarter,
       total_revenue,
       ROUND(avg_transaction_value, 2) AS avg_transaction_value, total_quantity_sold, total_transactions
FROM quarterly_sales
ORDER BY store_location, year, quarter;
```

- e) **Reference:** Extracted output has been visualized in fig 7 below.
- f) **Additional Notes:** Three parameters comparison across the store location including total quantity sold, total revenue and average transaction value has been calculated.

Section 2: Comparative Analysis of the KPIs

1. Fig 1. Shows Total Sales Per Quarter comparative analysis per store location –

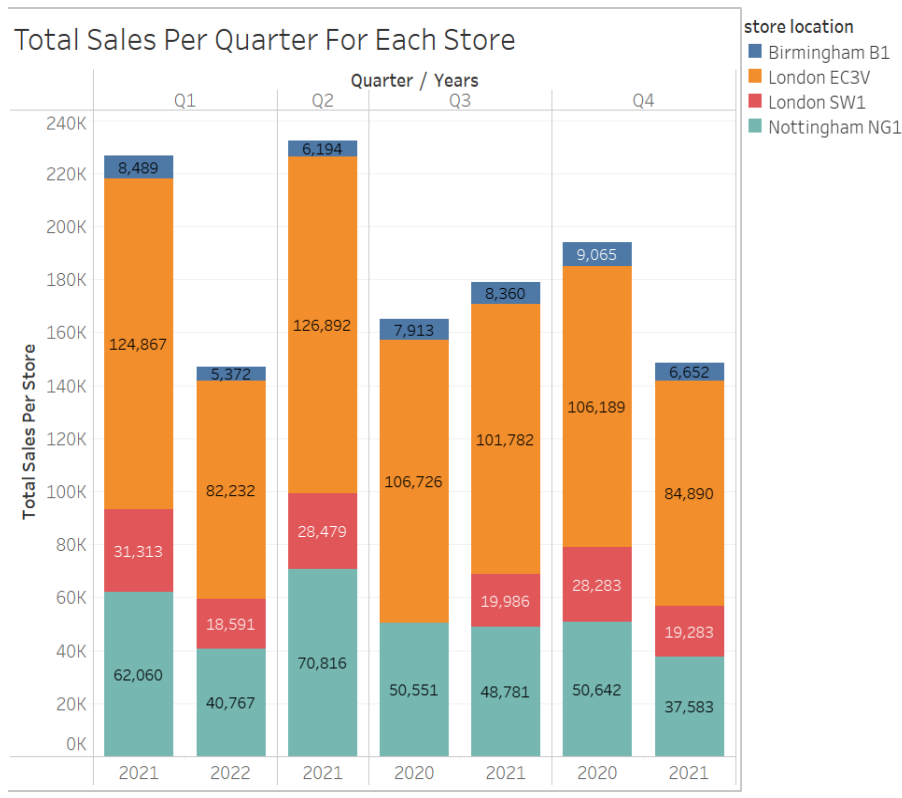


Figure 1: Total Sales Per Quarter for Each Store

- a) **London EC3V Dominates:** London EC3V consistently outperforms other locations in total quarterly sales across all years. Particularly strong in Q1 and Q2 of 2021 and 2022, indicating a stable customer base and strong demand.
- b) **London SW1's Moderate Performance:** London SW1 shows a steady contribution but lags behind London EC3V. Its sales peak in Q3 and Q4 of 2021, suggesting potential seasonal trends.
- c) **Nottingham NG1:** Nottingham NG1 has a fluctuating performance, with a significant sales drop in 2021 compared to 2020. The recovery in 2022 indicates potential for growth with the right strategies.
- d) **Birmingham B1's Underperformance:** Birmingham B1 records the lowest sales across all quarters and years. Consistently lower than other stores, indicating challenges in market penetration or customer retention.

2. Fig 2. illustrates The Sales Rate Per Day of Week comparative analysis per store location –

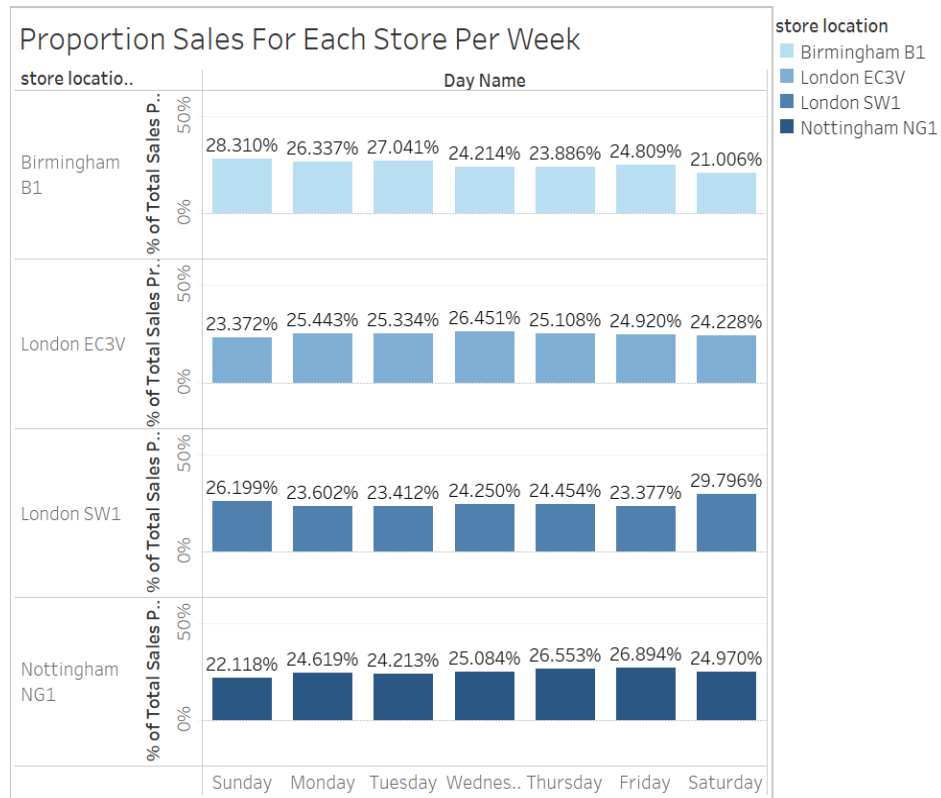


Figure 2: Proportion Sales for each store per weekdays.

- a) **Birmingham B1:** Sunday has the highest proportion of sales (28.31%), followed by Tuesday (27.04%). Saturday shows the lowest sales proportion (21.01%), indicating reduced customer activity on weekends compared to weekdays.
- b) **London EC3V:** Sales are more evenly distributed across the week, with a slight peak on Thursday (26.45%). Sunday (23.37%) has the lowest sales proportion, suggesting reduced activity.
- c) **London SW1:** Saturday (29.79%) shows the highest proportion of sales, unlike other locations, which suggests stronger weekend performance. Weekdays, especially Wednesday (23.41%), have relatively lower sales proportions.
- d) **Nottingham NG1:** Friday (26.89%) has the highest sales proportion, followed closely by Thursday (26.55%). Sunday (22.12%) has the lowest sales proportion, indicating reduced activity at the start of the week.

3. **Fig 3. illustrates Count of first-time customer in a specific period comparative analysis per store location –**

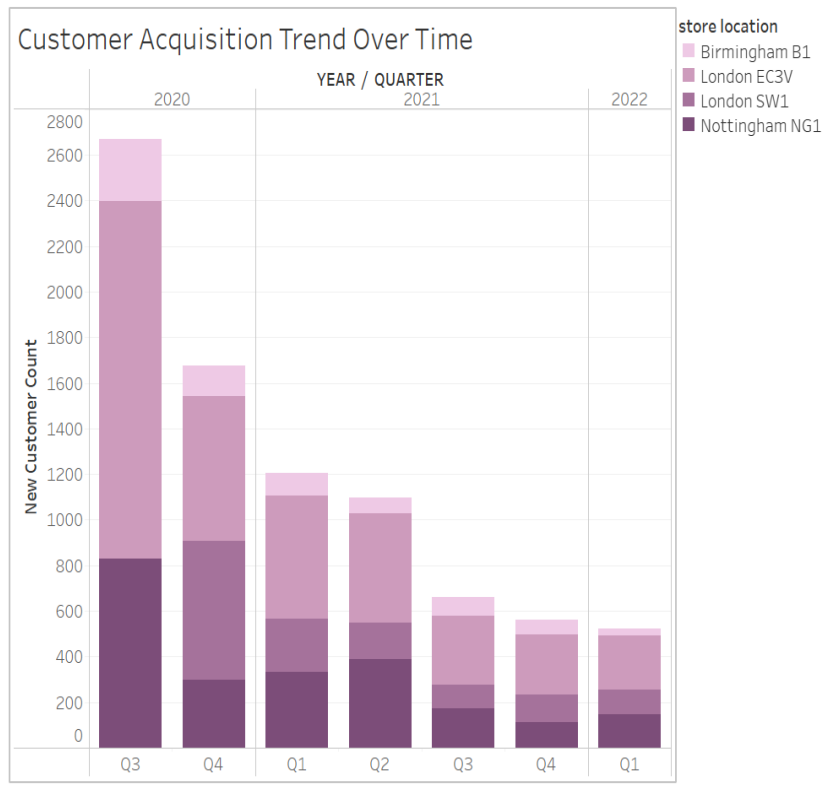


Figure 3 : Customer Acquisition Trend Over Time

- a) **2020 Peak:** The highest customer acquisition occurred in Q3 2020, with Birmingham B1 and London EC3V contributing significantly.
- b) **Decline Over Time:** There is a steady decline in customer acquisition from 2020 to 2022, across all locations.
- c) **Consistent Performance:** Despite the decline, London EC3V shows relatively consistent acquisition numbers compared to other locations.

4. **Fig 4. Represents customer lifetime value comparative analysis per store location –**

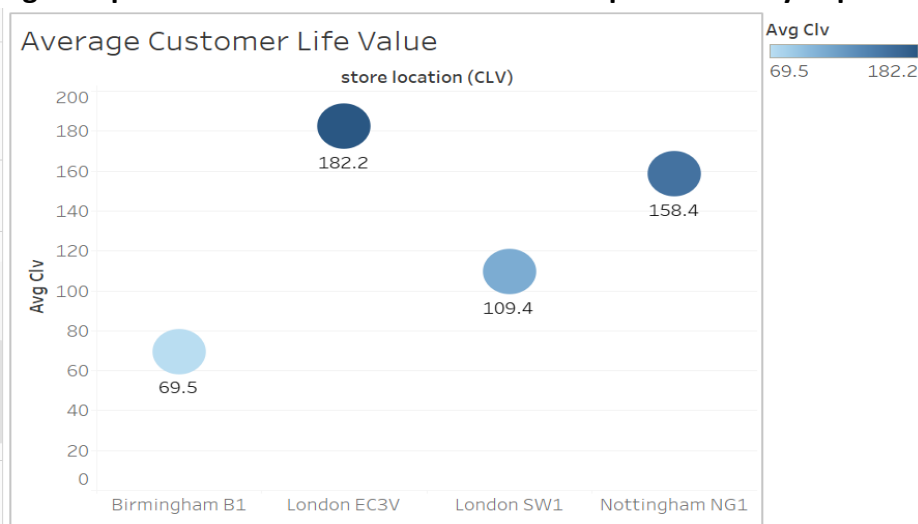


Figure 4 : Customer Lifetime Value Trend

- a) **Highest CLV:** London EC3V leads with the highest average CLV at 182.2, indicating a highly valuable customer base.
- b) **Lowest CLV:** Birmingham B1 has the lowest CLV at 69.5, reflecting less customer profitability.
- c) **Mid-performing Locations:** Nottingham NG1 (158.4) and London SW1 (109.4) have moderate CLV values, with Nottingham significantly outperforming London SW1.

5. Fig 5. Represents Cohort Analysis for Customer Retention Per Store Location –

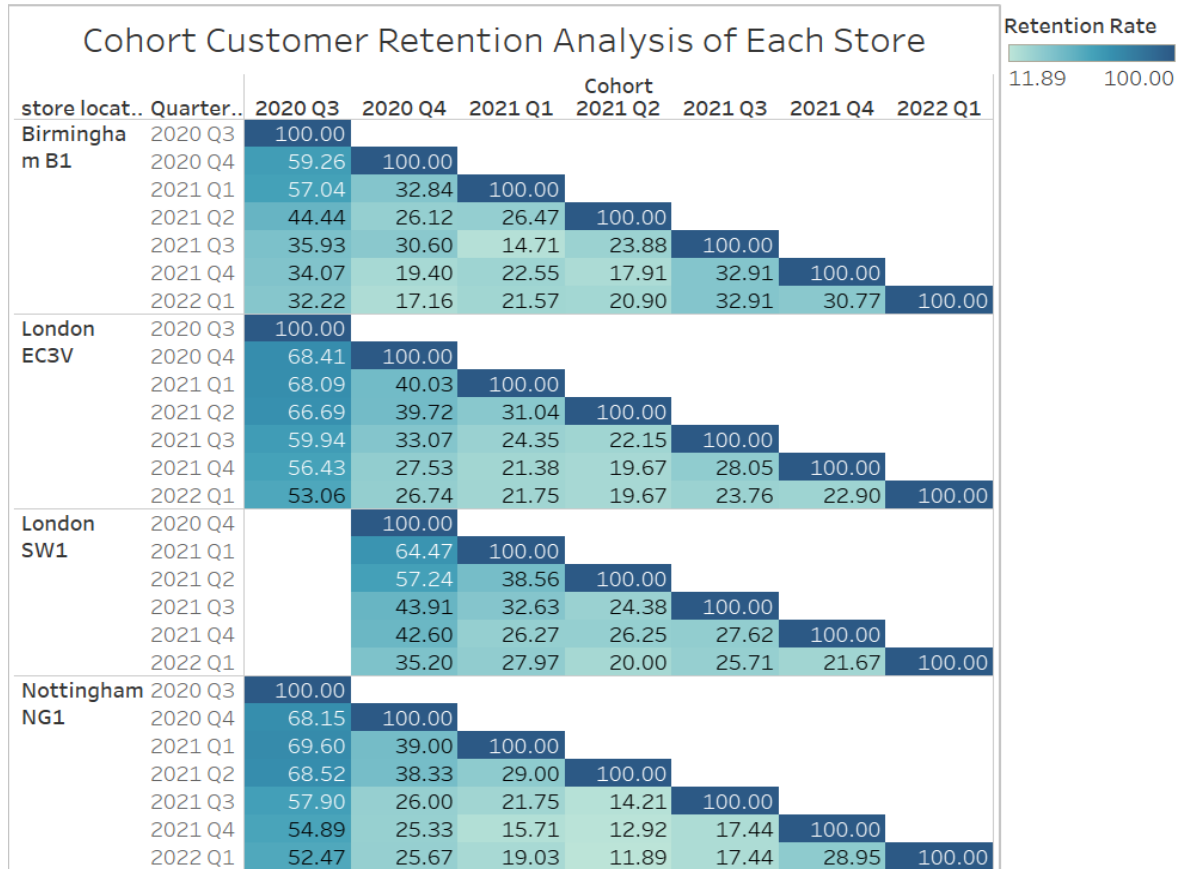


Figure 5: Cohort Customer Retention Per Store

- a) **Initial Retention (Quarter 0):** Each cohort starts with a 100% retention rate in the first quarter (since it includes all the customers who made a purchase in that cohort).
- b) **Retention Over Time:** Retention rates tend to decline over subsequent quarters, indicating customer churn. Some cohorts stabilize with certain retention levels, reflecting a loyal customer base that continues to make purchases.
- c) **Store Location Performance:**
 1. **Birmingham B1** shows relatively lower long-term retention compared to other locations.
 2. **Nottingham NG1** has higher initial retention rates across most cohorts, maintaining better performance in later quarters.
- d) **Retention Patterns:** In some cohorts, retention drops sharply in the first few quarters but stabilizes later (e.g., **London SW1, 2020 Q4** cohort). Certain cohorts demonstrate seasonal behaviour, where retention improves in specific quarters.

6. Fig 6. Shows Top five selling department analysis Per Store Location –

7.

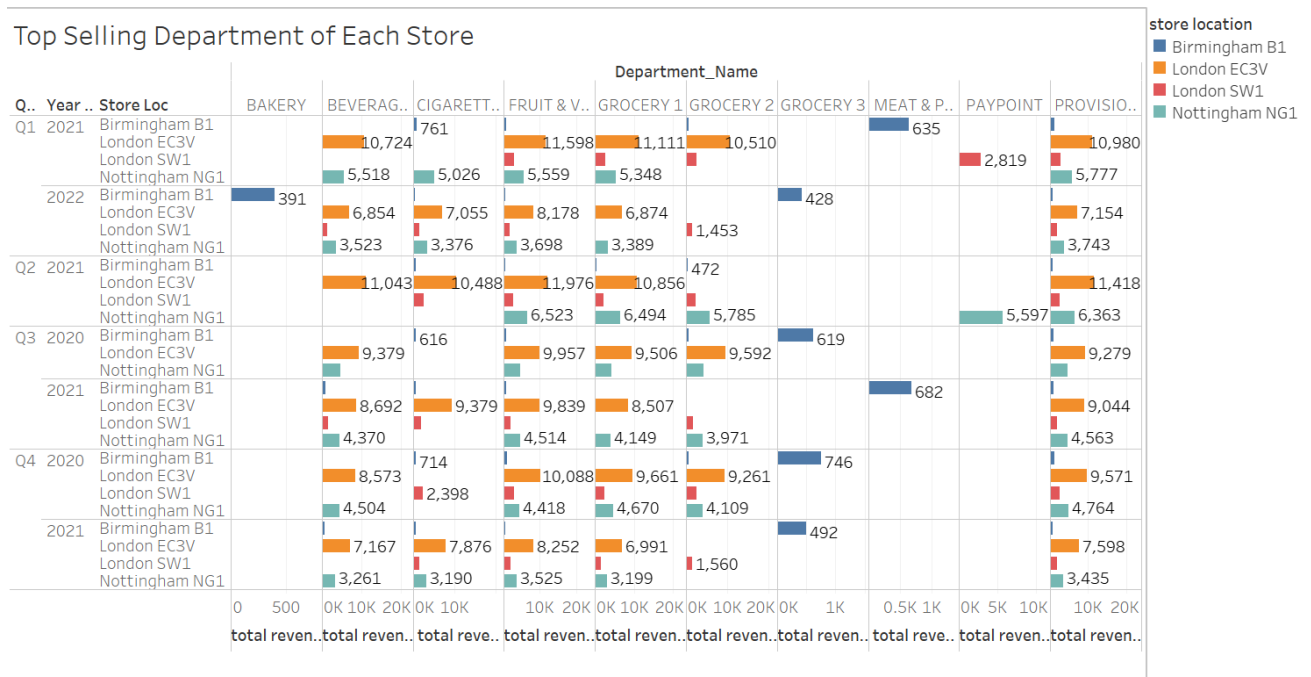


Figure 6 : Top Five Selling Department of each store

1. High-Performing Departments:

- London EC3V:** Dominates in "Grocery 1" and "Provision" departments consistently across quarters. Also performs well in "Fruit & Vegetables" and "Meat & Poultry" categories.
- Nottingham NG1:** Strong revenue from "Grocery 1" and "Provision" departments. "Fruit & Vegetables" consistently contribute significantly.
- Birmingham B1:** Limited performance across departments, with sporadic sales in "Meat & Poultry" and "Provision."
- London SW1:** Moderate performance in "Grocery 1," "Fruit & Vegetables," and "Provision". Sales vary significantly by department and quarter.

2. Underperforming Departments:

- "Cigarettes" and "PayPoint" generate low revenue across all stores.
- Birmingham B1 consistently underperforms across all departments, particularly in "Grocery" and "Provision."

3. Seasonal Trends:

- Revenue for "Provision" peaks in specific quarters (e.g., Q2 and Q4).
- "Fruit & Vegetables" and "Meat & Poultry" show stable demand across quarters, likely reflecting consistent customer needs.

8. Fig 7. Overall Store Performance analysis –

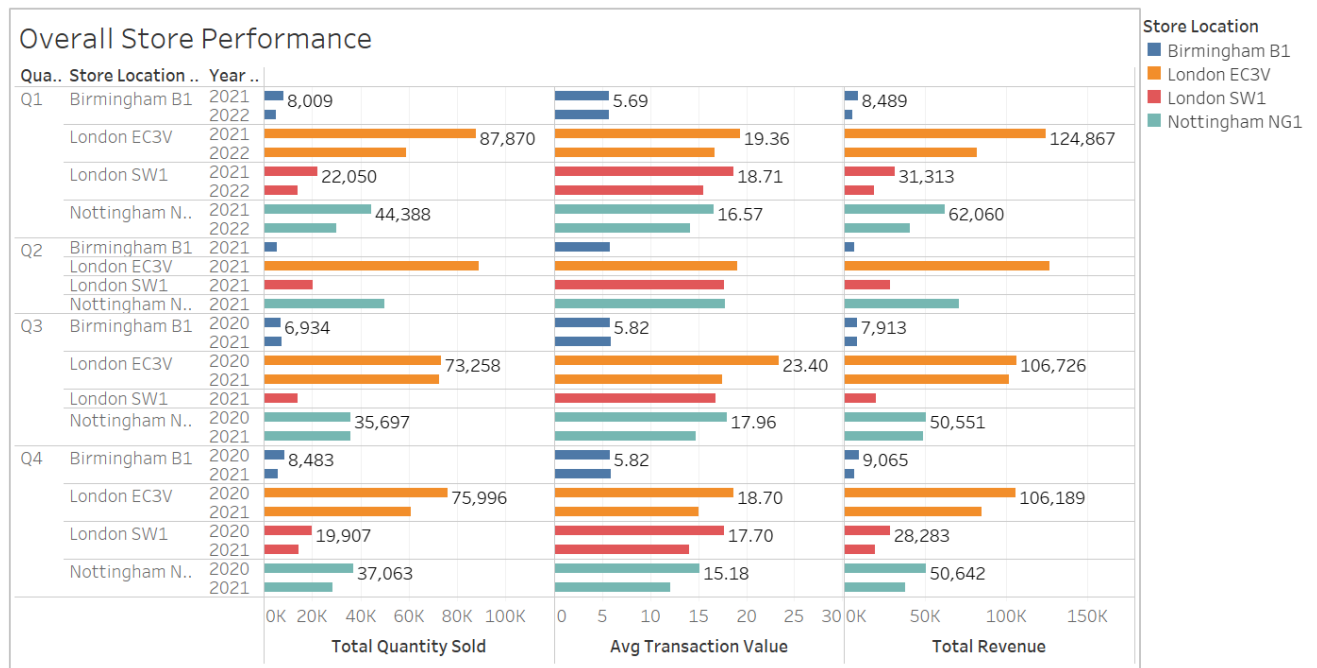


Figure 7: Overall Store Performance Trend

- Top Performer - London EC3V:** Consistently generates the highest total revenue and total quantity sold across all quarters and years. The store's performance significantly outpaces others, especially in 2021 and 2022.
- Nottingham NG1's Steady Growth:** Nottingham NG1 shows consistent performance with strong contributions to both total revenue and quantity sold. It ranked second across most quarters, suggesting a stable customer base.
- London SW1's Variability:** London SW1 has moderate but inconsistent performance, with revenue and sales fluctuating by quarter. Peaks observed in specific quarters, such as Q4 2020 and Q3 2021.
- Birmingham B1's Underperformance:** Records the lowest total revenue and quantity sold across all periods. A consistent struggle to compete with other stores in all quarters and years.

Business Recommendations:

1. Focus on High-Performing Stores:

London EC3V achieves the highest sales and revenue across all quarters consistently. This location should be prioritized for expansion or product experimentation.

2. Improve Retention in Birmingham and Nottingham:

The cohort analysis indicates lower customer retention rates in Birmingham B1 and Nottingham NG1 compared to other locations. Consider targeted retention strategies such as loyalty programs or personalized promotions.

3. Focus on High-CLV Locations & Revitalize Low-CLV Locations:

Prioritize marketing and customer retention efforts in **London EC3V** and **Nottingham NG1**, as they demonstrate strong customer profitability.

Investigate and address factors contributing to the low CLV in **Birmingham B1** through customer satisfaction surveys or targeted promotions.

4. Enhance Underperforming Departments:

Certain departments, such as **Bakery** and **Cigarettes**, have significantly lower revenue across most locations. Focus on improving product offerings or marketing for these departments.

5. Improve Birmingham B1's Performance:

Conduct market research to identify customer preferences and address gaps in product offerings. Launch aggressive promotions for underperforming departments to drive sales.

6. Seasonal Sales Optimization:

Q2 and Q4 consistently show higher sales across most stores. Use this trend to plan promotional campaigns and inventory stocking during these peak periods.

Conclusion:

London EC3V is a key revenue driver, while retention improvement in Birmingham and Nottingham can boost long-term performance. By focusing on strengthening weaker departments and taking advantage of seasonal trends, we can optimize profitability in all locations.